# Guy Wilson

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guyhwilson.com

## EDUCATION

**Stanford University**, Stanford, CA — **2018 – Present**
Ph.D. in Neuroscience (expected May 2023)

**University of California, Berkeley**, Berkeley, CA — **2014 – 2018**
B.A. in Mathematics, B.A. in Molecular & Cell Biology with Honors

## AWARDS & HONORS

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|---|---|
| BCI Society Student Award | 2021 |
| Mind, Brain, Computation, & Technology Student Membership | **2020** |
| National Science Foundation Graduate Research Fellowship | **2018 – 2021** |
| Regina Casper Stanford Graduate Fellowship | **2018 – 2023** |
| Phi Beta Kappa National Honor Society (offered) | **2018** |
| Departmental Honors (MCB) | **2018** |
| Distinction in General Scholarship | **2018** |

## EXPERIENCE

**Neurosciences Program, Stanford University**, Stanford, CA — **2018 – Present**
*Advisors: Krishna Shenoy & Shaul Druckmann*
- Developing BCI algorithms for speech decoding, cursor control, and unsupervised recalibration.

**BMI Software Intern, Neuralink**, CA — **Summer 2021**
*Supervisor: Joey O'Doherty*
- Machine learning stuff.
- Oversaw research projects end-to-end including problem identification, data collection, analysis, and codebase implementation.

**University of California, Berkeley**, Berkeley, CA
*Advisor: Robert Knight* — **Fall 2015 – 2018**
- Developed and coded analysis pipelines for human behavioral and ECoG data using dimensionality reduction, non/linear classifiers, and state-space methods.

*Psych 98/198: Neurotechnology Research Review Decal* — **Fall 2017**
- Designed and co-taught a new course on neurotechnology. Topics include statistical inference, fMRI, EEG/MEG, ECoG, TMS/tDCS, DBS, optogenetics, and in-vivo imaging.

## PUBLICATIONS

**GH Wilson,** FR Willett, E Stein, F Kamdar, DT Avansino, LR Hochberg, KV Shenoy, S Druckmann, JM Henderson (2023). Long-term unsupervised recalibration of cursor BCIs. (*in preparation*).

FR Willett*, E Kunz*, C Fan*, DT Avansino**, GH Wilson,** EY Choi, F Kamdar, LR Hochberg, S Druckmann, JM Henderson, KV Shenoy (2023). A high-performance speech neuroprosthesis. *bioRxiv*.

**GH Wilson*,** SD Stavisky*, FR Willett, DT Avansino, LR Hochberg, JM Henderson, S Druckmann, KV Shenoy (2020). Decoding spoken English from intracortical electrode arrays in dorsal precentral gyrus. *Journal of Neural Engineering*.

SD Stavisky, FR Willett, **GH Wilson**, B Murphy, P Rezaii, DT Avansino, et al. (2019). Neural ensemble dynamics in dorsal motor cortex during speech in people with paralysis. *eLife*.

RF Helfrich, M Huang, **GH Wilson**, RT Knight (2017). Prefrontal cortex modulates posterior alpha oscillations during top-down guided visual perception. *Proceedings of the National Academy of Sciences*.

## TALKS

**GRC Workshop: current advances in neurolinguistics**, Zurich, Switzerland — **Oct 2020**
Decoding spoken English from intracortical recordings in dorsal motor cortex. (Virtual)

**Stanford Computational Neuroscience Club**, Stanford, CA — **Nov 2019**
Linear dynamical systems for time-series data analysis.

## CONFERENCE ABSTRACTS

**GH Wilson,** SD Stavisky, D Avansino, LR Hochberg, JM Henderson, KV Shenoy, S Druckmann. Leveraging task structure for unsupervised recalibration of cursor BCIs. *Society for Neuroscience 2022.*

FR Willett*, E Kunz*, C Fan*, DT Avansino, **GH Wilson,** EY Choi, F Kamdar, LR Hochberg, S Druckmann, JM Henderson, KV Shenoy). A high-performance speech neuroprosthesis. *Society for Neuroscience 2022*.

**GH Wilson,** SD Stavisky, D Avansino, LR Hochberg, JM Henderson, KV Shenoy, S Druckmann. Decoding spoken English phonemes from dorsal motor cortex. *BCI Society International Meeting 2021, (virtual).*

SD Stavisky, **GH Wilson,** FR Willett, D Avansino, P Rezaii, LR Hochberg, S Druckmann, KV Shenoy, JM Henderson. Decoding speech production using intracortical electrode arrays in dorsal precentral gyrus. *BCI Unconference 2020, (virtual).*

**GH Wilson,** SD Stavisky, D Avansino, LR Hochberg, JM Henderson, KV Shenoy, S Druckmann. Neural state space geometry underlying speaking different phonemes. *Computational and Systems Neuroscience (COSYNE) 2020, Breckenridge, CO.*

SD Stavisky, **GH Wilson,** FR Willett, D Avansino, P Rezaii, LR Hochberg, S Druckmann, KV Shenoy, JM Henderson. Neural population dynamics in motor cortex during human speech. *CNEP UC Berkeley/San Francisco Annual Retreat 2019*.

**GH Wilson,** RF Helfrich, RT Knight. Spatiotemporal dynamics of contextual processing in human intracranial recordings. *MCB Honors Research Symposium 2018, Berkeley CA.*

M Huang, RF Helfrich, **GH Wilson,** RT Knight. Sequential and temporal predictions in the human brain employ neuronal alpha oscillations. *MCB Honors Research Symposium 2017, Berkeley CA.*

## OTHER PROJECTS

**Auditory-articulatory inversion model for speech decoding** — **2019**
*GH Wilson, S Druckmann*

- Built neural network models for predicting articulatory kinematics from audio waveforms.
- Experience with data cleaning, feature selection, and audio preprocessing approaches (frequency decompositions, voice conversion for data augmentation). Built in Python.

**Persistent homology of global functional connectivity for depression classification** — **2017**
*AR Hu, GH Wilson*

- Designed and implemented a topological data analysis pipeline for detection of clinical depression using resting-state fMRI.

**Recurrent neural networks for neoantigen prediction** — **2017**
*W Wang, L Huang, GH Wilson, J Price*

- Used a combination of existing genomics tools and structure-based machine learning to identify predict HLA-epitopes for cancer immunotherapy. Developed in Python for SVAI Genomics Hackathon.

## SKILLS & OTHER

**Programming:** Python, MATLAB, HTML, NumPy, Cython, PyTorch, GitHub.
Basic French, running, coffee.

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